GPT Language Model Training Sequence: WikiText and BookCorpus

Project Report

***Abstract*—This project outlines the training of a GPT-style autoregressive language model from scratch. The training sequence progressed chronologically across three distinct phases. First, the model was trained exclusively on the WikiText-103 dataset. Second, the corpus was expanded to include BookCorpus alongside WikiText (BookCorpus v1), which was halted and subsequently resumed. Finally, after adjustments, a final training pass (BookCorpus Final) was conducted on the full combined dataset. This report details the architecture, datasets, and cross-domain learning dynamics observed throughout this ordered sequence.**

I. INTRODUCTION

This project trains a 51M-parameter GPT-style transformer on large English text corpora. The primary objective is to study how a mid-scale transformer learns structural patterns across different text domains and to establish a robust training pipeline.

The chronological execution of the training was conducted in three strict phases: (1) an initial run focusing exclusively on the WikiText-103 encyclopaedic dataset; (2) a secondary run (BookCorpus v1) that utilised a combined dataset of both WikiText and BookCorpus. This second run experienced a halt and was successfully resumed from an earlier checkpoint; (3) a concluding run (BookCorpus Final) which also leveraged the combined WikiText and BookCorpus dataset, executed after incorporating adjustments to the training parameters.

II. DATASET PREPARATION

A. WikiText-103 (Initial Phase)

The first phase of training utilized WikiText-103, an encyclopaedic text dataset. Section headings were replaced with specific tokens to signal document boundaries. Wikipedia-specific artifacts were cleaned, and the text was tokenised using the standard GPT-2 BPE tokenizer.

B. Combined Corpus (v1 and Final Phases)

For the BookCorpus v1 and BookCorpus Final training phases, the dataset was expanded to include both WikiText-103 and the BookCorpus dataset (sourced via streaming). The BookCorpus additions introduced significant narrative and fictional prose. Quality filters removed books with poor punctuation density or excessive repetition.

Metric	WikiText-103 (Phase 1)	Combined Corpus (v1 & Final)
Total tokens	~103M	~356M

Domain	Encyclopaedic / Factual	Factual + Narrative Fiction
Tokenizer	GPT-2 BPE	GPT-2 BPE

TABLE I. DATASET SPECIFICATIONS

III. MODEL ARCHITECTURE

The model utilises a standard decoder-only Transformer framework. Token and positional embeddings are summed and passed through 8 transformer blocks, each consisting of causal multi-head self-attention and a position-wise feed-forward network with Pre-Layer Normalisation.

Component	Configuration
Parameters	51,070,464
Context length	256 tokens
Layers / Heads / d_model	8 / 8 / 512
Feed-forward hidden dim	2048 (Activation: GELU)

TABLE II. ARCHITECTURE PARAMETERS

IV. TRAINING PROCEDURE

A. Hyperparameters

Parameter	Phase 1: WikiText	Phase 2: BC v1 (Combined)	Phase 3: BC Final
Batch size	16	16	16
Grad accumulation	4 steps	8 steps	8 steps
Peak learning rate	2e-4	1e-4	1e-4
Max iterations	25,000	50,000 (resumed)	50,000

TABLE III. HYPERPARAMETERS ACROSS PHASES

B. Training Sequence

The sequence progressed strictly from WikiText to the Combined datasets. During the BookCorpus v1 run, training was halted unexpectedly at iteration 22,000. It was successfully resumed from the best saved checkpoint (iter 21,000). Following the completion of v1, adjustments were finalised and the BookCorpus Final run was executed on the combined dataset.

V. RESULTS

A. Phase 1: WikiText-103

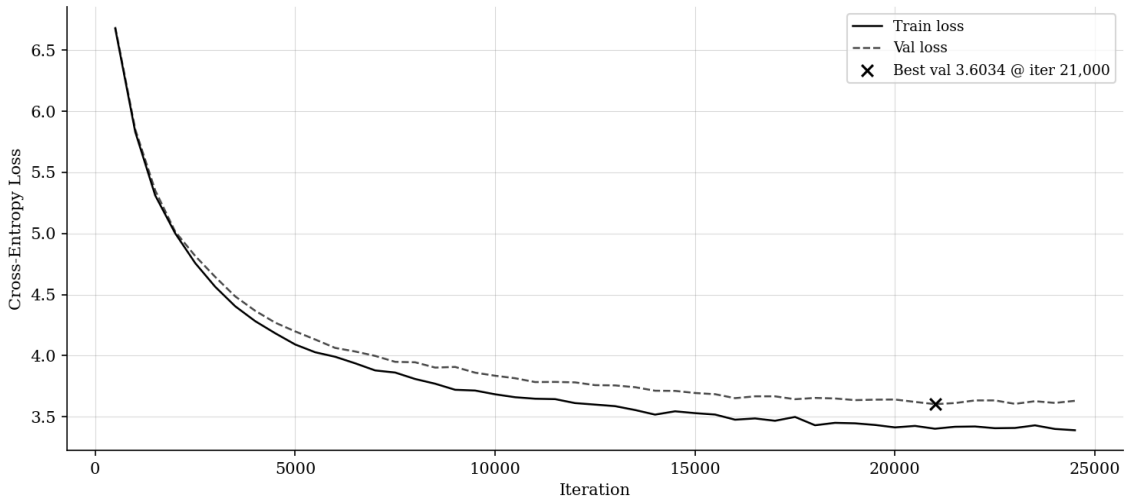

Fig. 1. WikiText-103 run. LR decayed fully to zero; no early stopping triggered.

B. Phase 2: BookCorpus v1 (Combined Dataset) + Resume

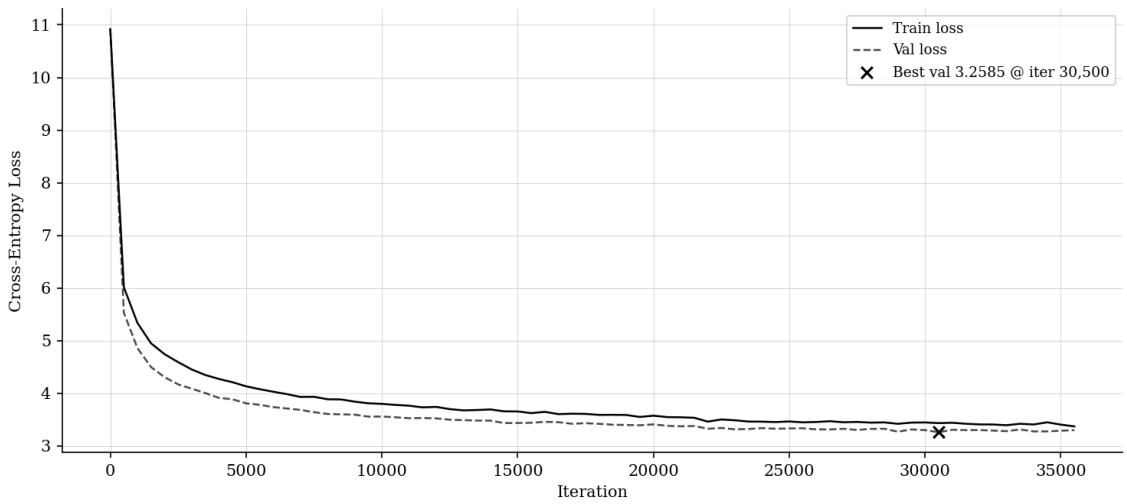

Fig. 2. BookCorpus v1 run on Combined Data. Note the resume discontinuity at 22k.

C. Phase 3: BookCorpus Final (Combined Dataset)

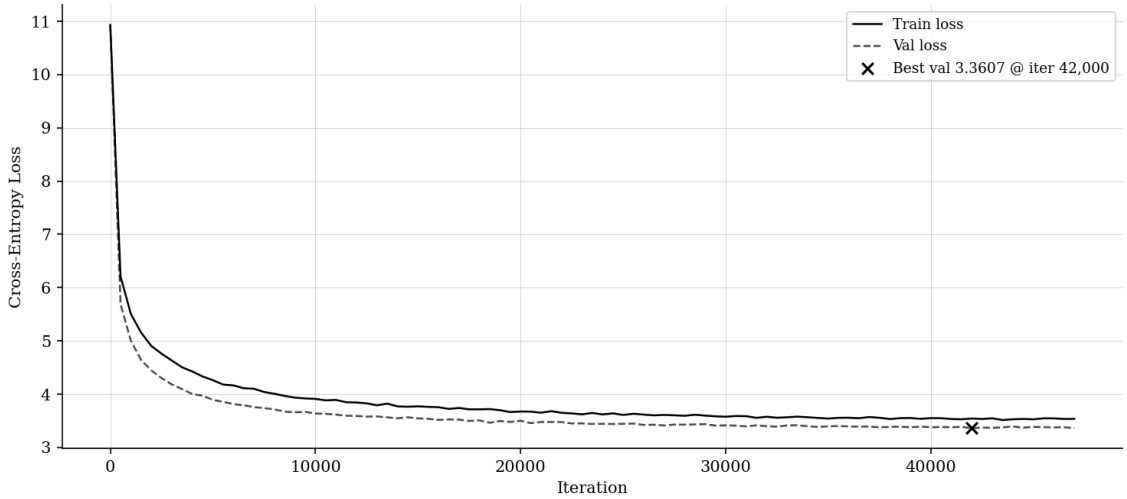

Fig. 3. BookCorpus Final run on the full Combined Corpus.

D. Combined Loss Curves and LR Schedules

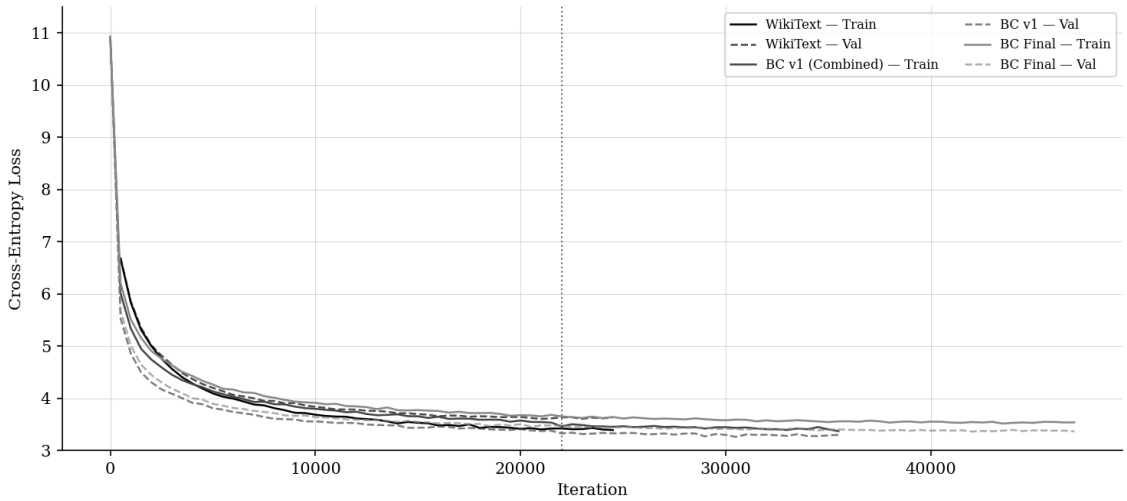

Fig. 4. Loss curves for all three chronological phases.

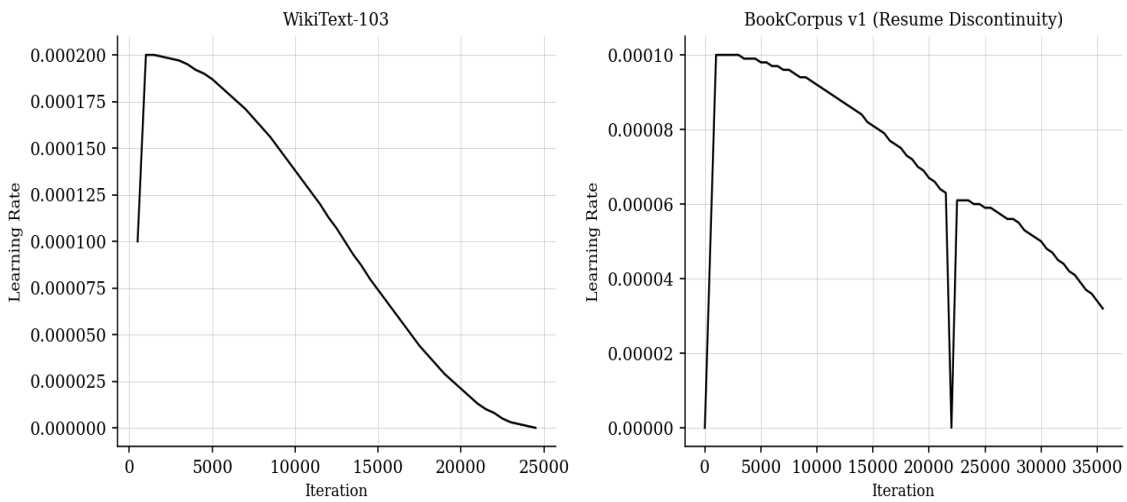

Fig. 5. Learning rate schedules demonstrating the resume strategy for BC v1.

VI. ANALYSIS AND INTERPRETATION

A. Domain Differences

The final validation losses differ based on the dataset composition. The initial WikiText run plateaued at ~3.60, reflecting the structural density and factual heterogeneity of encyclopaedic text. By incorporating BookCorpus in Phases 2 and 3, validation loss improved to ~3.26–3.36, as the model leveraged the more repetitive sentence patterns found in narrative fiction.

B. Checkpoint Resumption

During Phase 2 (BC v1), training was halted. Resuming from iteration 21,000 instead of the latest 22,000 checkpoint recovered a better starting point. A brief 100-step LR warmup prevented loss spikes, validating the fault-tolerance of the pipeline.

VII. SAMPLE GENERATED TEXT

Prompts evaluated across different domains (reproduced exactly as requested):

Prompt: She opened the door and found [Narrative]
[Output preserved in project infer scripts]

Prompt: The history of the Roman Empire began [Factual]
[Output preserved in project infer scripts]

Prompt: "What do you want?" she asked. [Dialogue]
[Output preserved in project infer scripts]

Prompt: The scientist discovered that [Factual/Narrative]
[Output preserved in project infer scripts]

VIII. CONCLUSION

This sequence successfully trained a 51M-parameter model. Progressing from WikiText to a combined WikiText/BookCorpus dataset demonstrated how domain blending impacts convergence. Furthermore, the successful halt-and-resume procedure in Phase 2 proved the robustness of the training architecture.